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PALAEOMAGNETISM, AGE, AND GEOCHEMISTRY OF THE DENCHAI BASALT, NORTHERN THAILAND

SANDRA M. BARR

Department of Geology, Acadia University, Wolfville, N.S. B0P 1X0 (Canada)

and

ALAN S. MACDONALD

R.R. 3, Wolfville, N.S. B0P 1X0 (Canada)

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The Denchai Basalt of northern Thailand is part of a large continental alkalic basaltic province extending through Thailand into Laos, Kampuchea, and Vietnam. It covers an area of about 70 km^2 and consists of seven flows, all with reversed magnetic polarity. The uppermost flow has yielded an age of 5.64 ± 0.28 Ma by the K-Ar method. During the igneous activity, magmas were apparently erupted from only one vent area, but changed in chemical composition from "hawaiite" (transitional to continental tholeiite) to true hawaiite and then to basanite. The basanite is the probable source of gem-quality zircon and sapphire mined from placer deposits associated with the basalts.

1. Introduction

Basalt flows cover an area of about 70 km^2 south of Denchai in northern Thailand (Fig. 1). These basalts are part of a large Late Cenozoic continental basaltic province extending through Thailand, Kampuchea, Laos, and Vietnam. Petrologic studies of basalt occurrences in Thailand have shown that these basalts are predominantly alkalic but exhibit considerable petrographic and geochemical variation [1]. Basalts farther to the southeast in Indochina show similar variations [2–5]. Basalts in this region of Southeast Asia are a major source of gem-quality zircon and corundum (sapphire and ruby). Reconnaissance geochemical data have suggested that gem-bearing basalts are generally more undersaturated than gem-free basalts [1,5].

Basalt flows in the Denchai area provide an excellent opportunity for a more detailed study of the stratigraphic and petrogenetic relationships of varied types of basalt in this igneous province, as previous reconnaissance sampling has indicated that the flows range from hawaiite to basanite [1]. Furthermore, zircon and sapphire are mined from gravels in valleys within and adjacent to the flows (Fig. 2).

Previously published studies of the Denchai Basalt have been limited to mapping of its distribution as part of a regional mapping programme [6] and reconnaissance palaeomagnetic measurements [7]. The latter study (based on three samples from one locality) indicated that the basalt has reversed magnetic polarity.

Although the basalt had not been previously dated, it has been assumed to be Late Cenozoic [6]. Dating of basalts elsewhere in Southeast Asia by K-Ar and fission-track methods has indicated a range of ages between 0.6 and 12.3 Ma [5,8–10]. Limited palaeomagnetic studies have indicated that basalt flows of both normal and reversed polarity occur in Indochina [3,4].

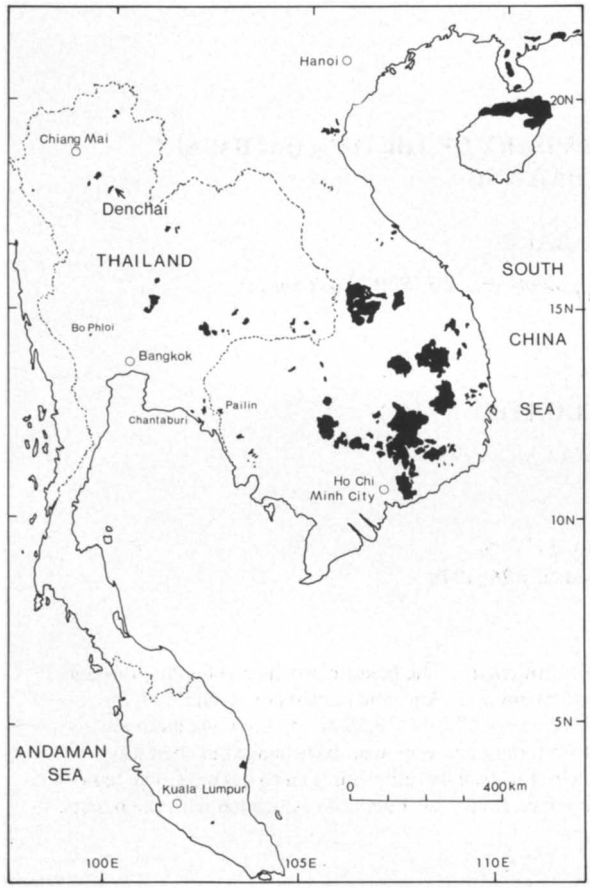


Fig. 1. Distribution of Late Cenozoic basalts in Southeast Asia. After Geological Map of Asia and the Far East, ECAFE, 1971.

2. Methods

A highway section through the basalt was mapped and sampled in detail during March 1976. Twenty-six oriented hand samples were collected from 19 localities. Two oriented cores for palaeomagnetic studies were later drilled from each sample in the Department of Geological Sciences, Chiang Mai University, Thailand. Thin sections were also prepared and 16 representative samples were selected for chemical analysis.

Silicon, aluminum, total iron, magnesium, calcium, titanium, and manganese were determined by atomic absorption spectrometry, sodium and potassium by flame photometry, and P_2O_5 by spectrophotometry in the Department of Geology, Acadia University. Analysis of H_2O^+ and H_2O^- followed the procedure of Peck [11] with minor modification. CO_2 was measured following the method of Shapiro and Brannock [12]. Ferrous iron was determined by titration with potassium permanganate solution to a visual end point. Ferric iron was determined by subtraction. All samples were analyzed in duplicate.

Trace element analyses were done using an INAX Instruments Energy Dispersive X-Ray Fluorescence analyzer at St. Mary's University (Halifax, Canada).

3. Geological setting and field relations

The Denchai Basalt occurs in an area of hilly terrain where elevations range from 200 to 500 m. The basalt follows a regional northeast-trending lineament (Fig. 2). The northern lobe of the basalt fills a broad valley with elevation 300–400 m. Gems are mined primarily from areas of low elevation in the eastern part of this valley near Ban Bo Keow and Ban Bo Soon. A topographic "high" (elevation 500 m) capped by basalt (which may be the vent area) separates this northern lobe from a more dissected southern lobe.

The country rocks are deformed sedimentary rocks of Permo-Carboniferous age [6]. At several localities, the basalt flows unconformably overlie baked argillites, locally separated by a thin layer of "caliche" (a palaeosol horizon?).

In the southern portion of the highway section, four flows with vesicular tops were clearly identified in the field (Fig. 2). From locality 6 to the northern limit of the basalt, at least three flows were identified on the basis of field observations. Although lack of exposure prevented definite stratigraphic correlations between these two groups of flows, the stratigraphic sequence shown in Fig. 2 was inferred on the basis of the field relations, petrography, and geochemistry (discussed below). Total exposed thickness of the flows is at least 25 m.

4. Palaeomagnetism and age

Intensities and directions of natural remanent magnetization (NRM) in the two cores drilled from each of the oriented hand samples were measured in the palaeomagnetic laboratory at Dalhousie University

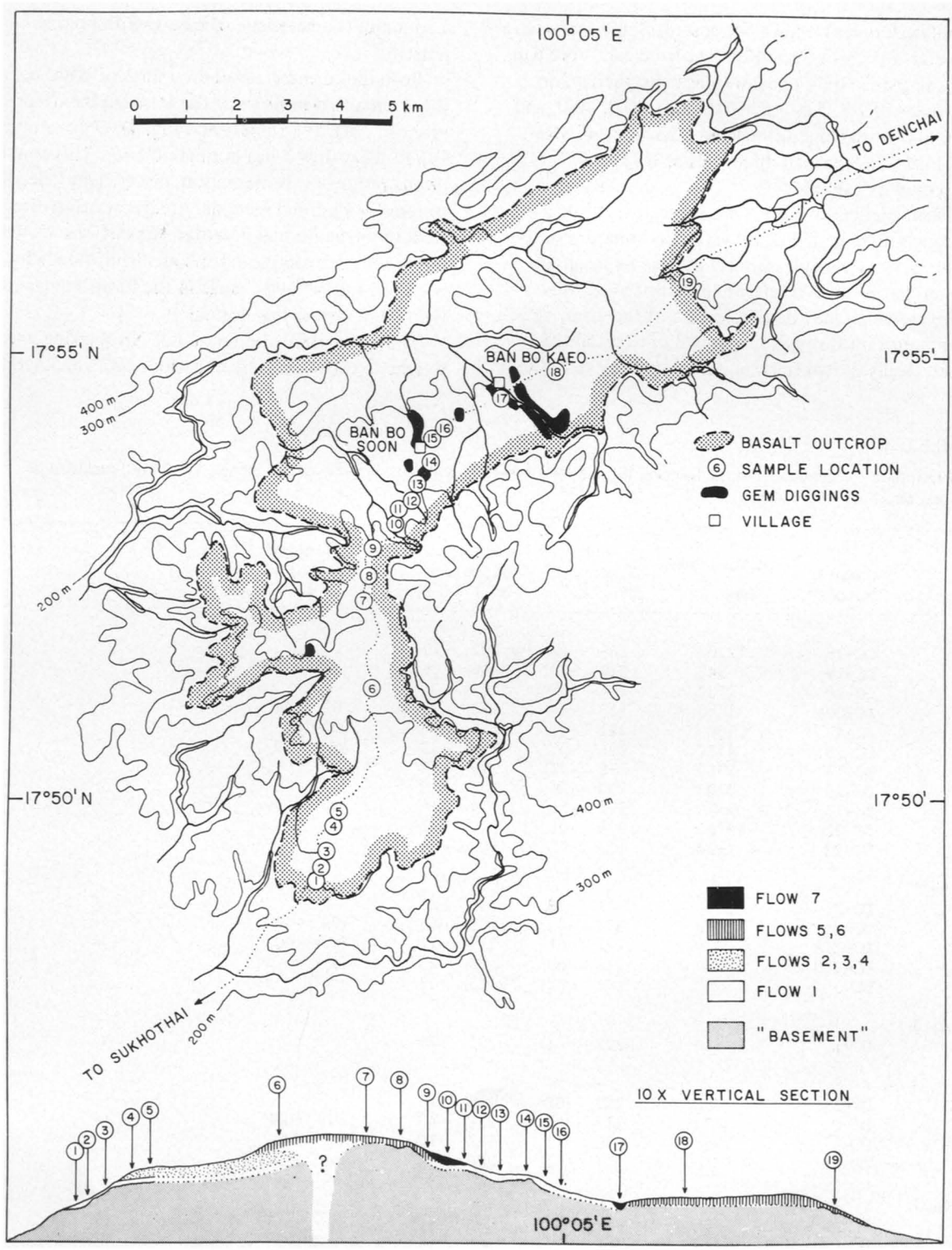


Fig. 2. Distribution of the Denchai Basalt (above) and interpretative cross-section along the Sukhothai–Denchai highway (below).

(Halifax, Canada) using a Schonstedt Spinner Magnetometer. In most cases NRM data from each core pair were similar so only one core was demagnetized in a.c. fields of 2.5, 5.0, 7.5, 10.0, 20.0, 30.0, 40.0, and 50.0 mT. Initial and normalized intensities of NRM, and initial and cleaned directions of NRM are presented in Table 1.

Demagnetization curves are generally smooth although there is significant variation in hardness (Fig. 3a, b, c). Samples from flow 7 show magnetization that is much harder than that of most other samples, which may be a reflection of the glassy texture and/or uniformly fine-grained opaques in this basalt. Many curves show an initial increase in NRM, a common characteristic of basalts with reversed polarity.

Both initial and cleaned directions of NRM confirm that all seven flows of the Denchai Basalt have reversed polarity (Table 1 and Fig. 4). Only sample DC-24 from flow 7 has normal polarity. This could be the result of misorientation, insufficient "cleaning" to remove a normal secondary magnetization direction, or a true normal polarity. The last case would imply that this sample is from an eighth flow which occurred following a change in the Earth's magnetic field from reversed to normal.

The mean cleaned direction of NRM giving unit weight to each lava is: declination, 200°; inclination,

TABLE 1

Magnetic data for samples from the Denchai Basalt. NRM = natural remanent magnetization. Sample DC-24 not included in calculation of mean direction

Locality	Sample Number	Initial NRM (mAm^{-1})	NRM Direction D(°)	NRM Direction I(°)	Cleaned Direction D(°)	Cleaned Direction I(°)	Mean Cleaned Direction D(°)	Mean Cleaned Direction I(°)
Flow 1								
1	DC-18	11100	169	-12	192	-19		
2	DC-19	740	229	-29	216	-36		
3	DC-20	1730	204	-35	207	-31		
3	DC-20A	1750	207	-44	190	-38	201	-34
4	DC-4	1000	221	-33	215	-35	(κ = 24)	
11	DC-13	4720	196	-33	197	-33	(α_{95} = 9°)	
12	DC-14	2560	198	-46	196	-48		
13	DC-15	3100	190	-26	187	-28		
14	DC-16	3510	172	-26	174	-26		
15	DC-22	18700	339	-06	241	-32		
16	DC-23	4130	179	-32	183	-34		
Flow 2								
3	DC-2	9330	311	-08	198	-29		
3	DC-21	8600	303	-01	205	-29	190	-28
3	DC-21A	830	222	-01	211	-11	(κ = 39)	
3	DC-21B	4370	116	-09	180	-25	(α_{95} = 16°)	
4	DC-3	990	183	-32	183	-32		
Flow 4								
5	DC-6	610	202	-21	201	-29		
Flow 5 or 6								
6	DC-1	890	259	-08	209	-26		
7	DC-8	3470	217	-41	218	-43	201	-24
9	DC-9	7750	076	-11	181	-10	(κ = 16)	
18	DC-25	6660	217	+01	200	-14	(α_{95} = 18°)	
Flow 7								
9	DC-10	5500	219	-37	219	-40	207	-33
10	DC-12	4190	195	-24	196	-26	(κ = 148)	
17	DC-24	4270	002	+44	006	+42	(α_{95} = 20°)	

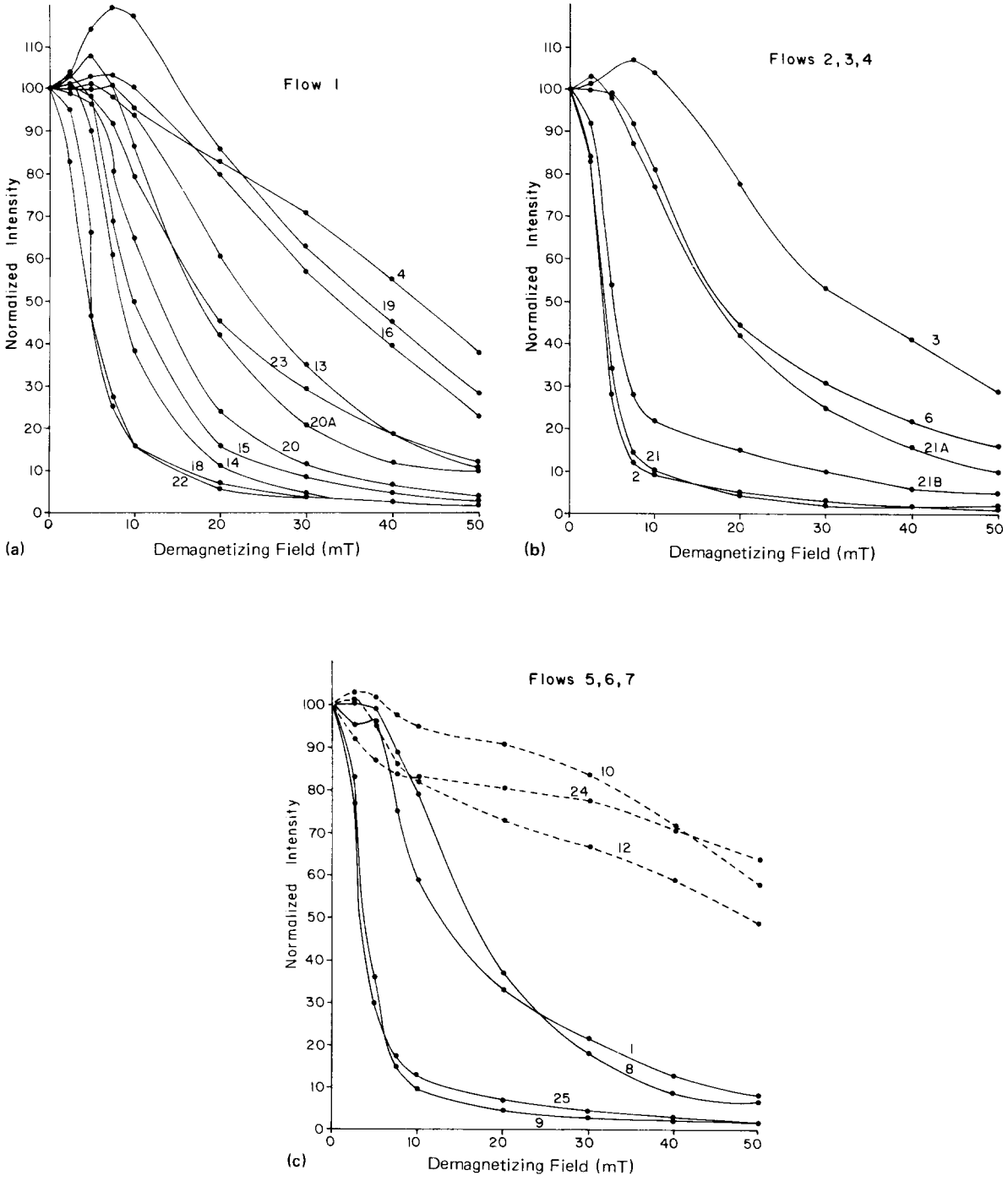


Fig. 3. Demagnetization curves for samples from the Denchai Basalt. (a) 11 samples from flow 1; (b) 6 samples from flows 2, 3, and 4; (c) 4 samples from flows 5 and 6 (solid lines); and 3 samples from flow 7 (dashed lines).

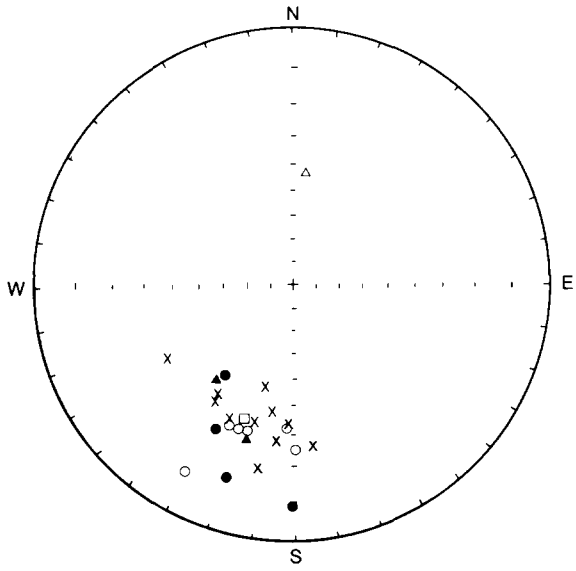


Fig. 4. Cleaned directions of magnetization for 24 samples from the Denchai Basalt plotted on a Wulff stereonet. Flow 1, crosses; flows 2, 3, and 4, open circles; flows 5 and 6, closed circles; flow 7, closed triangles; sample DC-24, open triangle; mean direction, square.

$-30°$ ($k = 148$, $\alpha_{95} = 5°$). This probably does not differ significantly from the present magnetic field direction (reversed declination, 180°; reversed inclination, $-20°$) when effects of secular variation are considered.

A sample of basalt from flow 7 (locality 8) has been dated by the K-Ar method in the Institute of Geological Sciences (London, Great Britain) and yielded an age of 5.64 ± 0.28 Ma [13]. This age may correspond to the period of reversed polarity which occurred between 6.06 and 5.62 Ma [14]. As this age date came from the uppermost flow [7], the underlying flows (all having reversed polarity) could have been extruded during this time interval. Alternatively, they could have occurred during an older reversed polarity epoch.

5. Petrology and geochemistry

Flow 1. This flow, apparently the most extensive of the sequence, unconformably overlies baked argillites at several localities within both the northern and southern lobes (Fig. 2). Where exposed (localities 4 and 11), the top of the flow is highly vesicular. Maximum observed thickness is about 5 m.

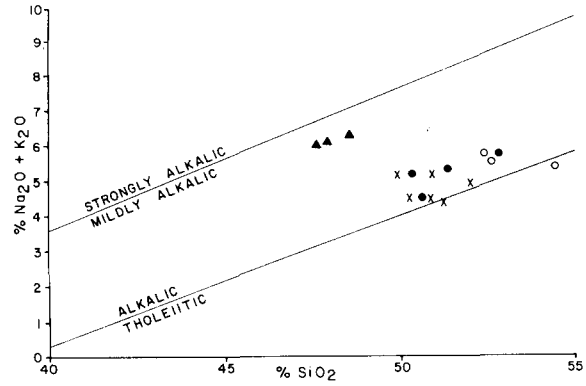


Fig. 5. Alkalies-silica diagram for the Denchai Basalt. Symbols as in Fig. 4.

In thin section the basalt is fine-grained to medium-grained, with plagioclase (andesine-labradorite), clinopyroxene, and opaque minerals in intergranular texture. Phenocrysts and microphenocrysts of olivine (Fo_{75-80}) are generally abundant, together with minor plagioclase (labradorite) and rarely clinopyroxene. The olivine crystals show varying degrees of alteration to iddingsite. Zeolites and alteration minerals are abundant in some samples.

Chemical analyses confirm the distribution of flow 1 deduced from field relations; the six samples from this flow show close chemical similarity (Table 2).

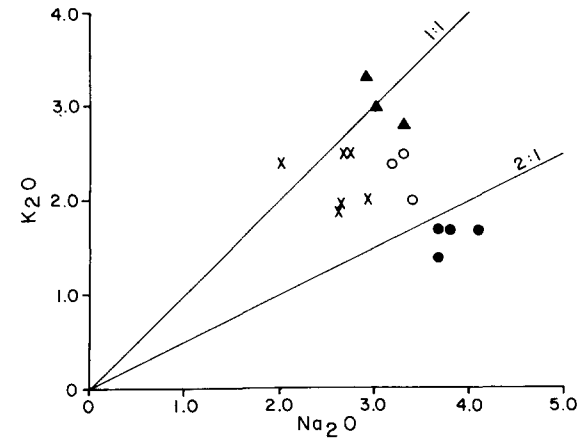


Fig. 6. Plot of K_2O against Na_2O for the Denchai Basalt. Symbols as in Fig. 4.

TABLE 2

Chemical data and CIPW normative mineralogies for the Denchai Basalt. Oxides in weight percent, elements in ppm. Norms were calculated on a volatile-free basis using the following standard ratios for Fe_2O_3/FeO [1]: flows 1–4, 0.25; flows 5, 6, 0.40; flow 7, 0.20

	FLOW 1						FLOW 2	FLOW 3	FLOW 4	FLOWS 5 and 6				FLOW 7		
Locality	12 DC-14	13 DC-15	1 DC-18	3 DC-20	15 DC-22	16 DC-23	4 DC-3	5 DC-5	5 DC-6	6 DC-1	18 DC-25	7 DC-7	9 DC-9	9 DC-10	10 DC-12	17 DC-24
SiO_2	47.8	47.6	50.9	49.3	49.5	49.8	50.9	50.9	51.2	50.3	49.1	47.5	49.1	45.9	45.7	46.4
TiO_2	1.86	1.8	1.91	1.84	1.87	1.88	1.7	1.7	1.67	1.67	1.97	1.77	1.85	2.1	2.26	2.16
Al_2O_3	14.7	15.7	14.8	14.8	15.3	14.7	16.3	16.5	14.4	14.6	15.7	14.5	14.7	15.7	14.6	15.2
Fe_2O_3	2.9	1.7	1.6	2.2	2.1	2.2	2.9	3.8	3.3	2.7	2.3	3.1	2.8	2.0	4.3	2.5
FeO	6.4	8.2	8.5	7.1	7.2	7.2	5.3	4.5	4.5	5.7	7.2	6.5	6.9	7.5	5.7	6.8
MnO	0.16	0.15	0.17	0.15	0.18	0.16	0.10	0.12	0.11	0.15	0.16	0.15	0.15	0.14	0.18	0.18
MgO	8.7	8.4	7.2	8.3	8.6	8.5	6.8	6.4	6.0	6.8	7.7	7.6	7.3	8.4	9.4	7.1
CaO	7.7	8.6	7.6	7.6	7.3	8.4	6.9	7.0	7.4	7.3	7.8	7.4	7.4	8.2	7.8	8.8
Na_2O	2.5	2.3	2.8	2.6	1.9	2.5	3.1	3.2	3.2	3.9	3.6	3.6	3.5	3.4	2.9	2.8
K_2O	1.8	1.9	2.0	2.4	2.3	1.9	2.3	2.4	1.9	1.6	1.4	1.6	1.6	2.5	2.9	3.2
P_2O_5	0.44	0.50	0.39	0.42	0.46	0.41	0.50	0.50	0.47	0.51	0.50	0.30	0.36	0.55	0.61	0.69
H_2O^+	2.6	2.5	0.9	1.8	1.5	2.1	2.2	1.6	0.9	1.7	2.1	2.9	2.3	2.1	1.6	1.8
H_2O^-	1.9	0.4	0.9	1.2	1.4	1.0	1.4	2.1	2.3	1.2	1.2	1.4	1.2	1.0	2.0	1.5
CO_2	0.4	0.2	0.2	0.2	0.2	0.2	0.1	-	0.5	0.7	0.3	0.4	0.3	0.2	0.3	0.9
Total	99.9	99.9	99.9	99.9	99.8	100.7	100.5	100.7	97.9	98.8	101.0	98.7	99.5	99.7	100.3	100.0
Q	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.7	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Or	11.2	11.6	12.1	14.7	14.1	11.5	14.1	14.7	12.0	9.9	8.5	10.1	9.9	15.4	17.9	19.8
Ab	22.3	20.1	24.2	22.8	16.6	21.6	27.1	28.0	28.8	34.7	31.2	31.9	31.0	15.9	14.6	14.7
An	24.8	27.8	22.4	22.4	27.3	23.8	24.6	24.3	20.6	18.5	23.1	19.9	20.6	21.0	19.0	20.3
Ne	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.0	7.6	5.9	5.5
Di	9.9	10.5	11.0	11.1	5.7	13.4	6.0	6.6	12.3	12.9	10.5	13.8	12.4	14.2	13.8	16.7
Hy	16.0	9.5	20.5	11.3	23.0	15.4	17.2	14.1	17.8	9.0	5.3	0.0	8.2	0.0	0.0	0.0
Ol	6.9	13.3	2.2	10.3	0.9	7.0	4.0	5.4	0.0	6.8	12.3	15.6	9.2	18.2	20.4	14.9
Mt	4.1	2.5	3.1	2.9	2.9	2.8	2.4	2.4	2.4	3.7	4.0	4.2	4.2	2.4	2.4	2.3
Il	3.7	3.5	3.7	3.6	3.7	3.7	3.3	3.3	3.4	3.3	3.8	3.6	3.7	4.1	4.5	4.3
Ap	1.1	1.2	0.9	1.0	1.1	1.0	1.2	1.2	1.2	1.2	1.2	0.7	0.9	1.3	1.5	1.7
Total	100.1	100.0	100.1	100.1	100.3	100.2	100.3	100.0	100.2	100.0	99.9	100.1	100.1	100.1	100.0	100.2
Rb	23	32	34	44	34	27	-	46	13	41	24	56	28	30	40	33
Sr	811	555	569	692	573	535	-	549	596	642	588	442	971	659	692	778
Nb	30	37	36	30	30	31	-	35	30	25	36	24	26	40	44	42
Zr	189	169	189	177	173	164	-	179	182	178	190	138	163	197	208	227
Ba	306	287	267	291	302	276	-	362	408	466	381	254	276	376	399	433
La	18	17	17	19	17	17	-	19	20	25	23	14	17	25	24	27
Ce	34	32	27	33	34	34	-	40	39	42	44	27	31	43	45	50
Zr/Nb	6.3	4.6	5.3	5.9	5.8	5.3	-	5.1	6.1	7.1	5.3	5.8	5.8	4.9	4.7	5.4
Zr/La	10.5	9.9	11.1	9.3	10.2	9.7	-	9.4	9.1	7.1	8.3	9.9	9.6	7.9	8.7	8.4

The average analysis is given in Table 3. In some respects the basalt resembles typical continental tholeiite, but low iron and high alkalies suggest alkalic affinities (Fig. 5) and it is classified as "transitional hawaiite". Potash is relatively high, with Na_2O/K_2O averaging 1.2 (Fig. 6). The normative plagioclase composition is about An_{50}, and the samples contain moderate to large amounts of normative hypersthene (Fig. 7).

Flows 2, 3, and 4. These three flows are chemically and petrographically similar, and hence are grouped for purposes of discussion. Contacts between flows are sharp, each flow having a vesicular top, and no erosion is indicated between these flows or between flows 1 and 2. The thickness of each flow is approximately 2–3 m.

In thin section, these basalts consist of fine- to medium-grained plagioclase (andesine), fine-grained granular clinopyroxene, and needles of opaque minerals in intergranular texture. Abundant olivine phenocrysts (Fo_{65-80}), scattered plagioclase phenocrysts (labradorite) and rare clusters of clinopyroxene crystals are present. Alteration and oxidation have occurred to varying extents.

Chemically these flows differ significantly from flow 1 (Tables 2, 3). Like flow 1, they are mildly alkaline (Fig. 5) but contain higher silica and soda

TABLE 3

Average chemical data for flows of the Denchai Basalt. Average alkali olivine basalt and continental tholeiite from Hyndman [15]

	Flow 1	Flows 2,3,4	Flows 5,6	Flow 7	Average Alkali Olivine Basalt	Average Continental Tholeiite
SiO_2	50.8	53.2	51.1	47.9	47.1	50.7
TiO_2	1.9	1.8	1.9	2.3	2.7	2.0
Al_2O_3	15.5	16.4	15.6	15.8	15.3	14.4
FeO^T	9.7	8.1	9.4	9.7	12.2	12.7
MnO	0.16	0.11	0.15	0.17	0.17	0.20
MgO	8.5	6.7	7.7	8.6	7.0	6.2
CaO	8.1	7.4	7.8	8.6	9.0	9.4
Na_2O	2.5	3.3	3.8	3.2	3.4	2.6
K_2O	2.1	2.3	1.6	3.0	1.2	1.0
P_2O_5	0.45	0.51	0.44	0.64	0.41	-
Na_2O/K_2O	1.2	1.4	2.4	1.1	2.8	2.6
Rb	32	30	37	34		
Sr	623	573	661	710		
Nb	32	33	28	42		
Zr	177	181	167	211		
Ba	288	385	344	403		
La	18	20	20	25		
Ce	32	40	36	46		
Zr/Nb	5.5	5.5	6.0	5.0		
Zr/La	9.8	9.1	8.4	8.4		

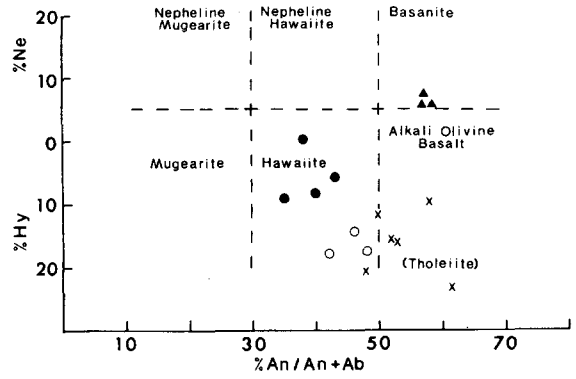


Fig. 7. Plot of normative hypersthene or nepheline against normative plagioclase composition of the Denchai Basalt. Symbols as in Fig. 4. Field boundaries from Barr and Macdonald [1].

with average $Na_2O/K_2O = 1.4$ (Fig. 6). The samples contain large amounts of normative hypersthene (Fig. 7). Silica variation diagrams suggest that magma of flows 2, 3, and 4 may have been produced by differentation of flow 1 magma (Fig. 8). These basalts are classified as "evolved transitional hawaiites".

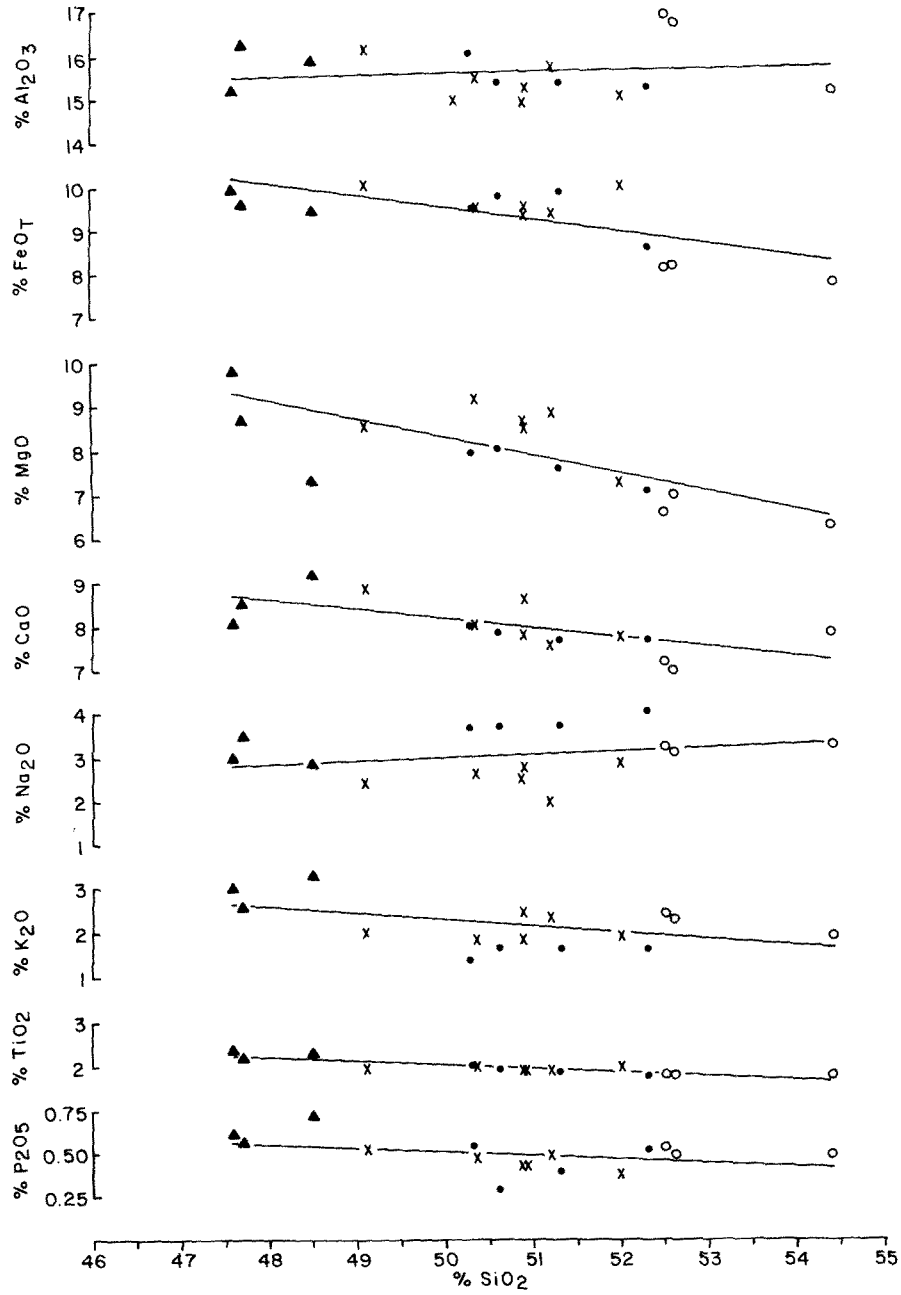


Fig. 8. Silica variation diagrams for the Denchai Basalt. Symbols as in Fig. 4. Best-fit lines determined by linear regression.

Flows 5 and 6. Both flows 5 and 6 are exposed at locality 7 (Fig. 2) where the vesicular top of flow 5 is overlain by massive nodule-bearing basalt of flow 6. The two flows have a combined thickness of at least 8 m. The ultramafic nodules are concentrated in a layer about 1 m in thickness near the bottom of flow 6. Basalts at localities 6 and 18 are tentatively correlated with these flows on the basis of geochemistry (Table 2).

In thin section, these basalts have fine-grained to

medium-grained intergranular textures with abundant olivine phenocrysts and microphenocrysts (Fo_{75-80}). Plagioclase phenocrysts occur much less abundantly. The intergranular groundmass contains zoned plagioclase (An_{35-50}), small grains of pyroxene, and large abundant opaques. The ultramafic nodules are generally spinel lherzolite.

Chemically these basalts are mildly alkaline (Fig. 5) with high soda and low potash compared to underlying flows (average $Na_2O/K_2O = 2.4$). They are classified as hawaiites (Fig. 7).

Flow 7. This distinctive flow occurs at localities 10–12 (overlying flow 1), at locality 9 (overlying flow 6), and at locality 17 where it is the only flow exposed (Fig. 2). At locality 10, flows 1 and 7 are separated by an erosional horizon consisting of baked gravels and soil containing basaltic boulders. Flow 7 is at least 3–4 m in thickness.

Basalt of flow 7 is aphanitic, black, and massive, and outcrops show good columnar jointing. It contains abundant ultramafic nodules (spinel lherzolite) up to 10 cm or more in diameter.

In thin section the basalt consists of a reddish-brown glassy matrix, with brown clinopyroxene crystals, scattered small plagioclase laths (An_{40-50}), abundant olivine phenocrysts (Fo_{82}) and abundant finely disseminated opaques. Plagioclase laths show flow alignment.

Chemically, the flow is distinctly different from underlying flows (Tables 2 and 3). It is more strongly alkaline (Fig. 5) and higher in potash, with average $Na_2O/K_2O = 1.1$ (Fig. 6). It is more undersaturated than the other flows, containing normative nepheline, and is classified as basanite (Fig. 7).

6. Source of gems

Sapphire and zircon are mined from "gravel" deposits in several areas on the Denchai Basalt (Fig. 2). Unfortunately these minerals have not been observed in the basalt itself so it is not possible to state with certainty which flow or flows have yielded these gem minerals. However the known gem occurrences cluster in the northern part of the basalt, in the area where flows 5, 6 and 7 are exposed on high ground, and these flows seem the most likely sources.

Elsewhere in Southeast Asia, only one type of basalt is generally found in important gem-producing regions. In the Chantaburi and Pailin areas (Fig. 1) very undersaturated basanites and nephelinites are the source of sapphire, ruby, zircon, and other gem minerals, whereas at Bo Phloi, nepheline hawaiite is the source of primarily zircon and spinel [1]. Most other basalt occurrences in Southeast Asia are not known to be gem-bearing, and are dominantly hawaiitic or tholeiitic [6].

On the basis of geochemistry then, flow 7 seems to be the most likely source of the gems. Like gem-bearing flows in the Bo Phloi, Chantaburi and Pailin areas, flow 7 contains abundant ultramafic nodules, whose presence and petrography suggest rapid ascent of magma from deep in the mantle.

The data of the present study do not contribute new insight towards an understanding of the ultimate origin of the gems, the subject of considerable speculation and debate [1,3,16–18]. However, they support previous observations [1] of the distinctive petrography and geochemistry of the gem-bearing basalts in Southeast Asia, significant evidence toward future understanding of the origin of the gem minerals.

7. Petrogenesis of Denchai Basalt

Alkaline basalt magmas are generally agreed to have originated by partial melting of the mantle at relatively deep levels. A variety of alkaline magmas may be produced by variations in the amount and depth of both partial melting and fractional crystallization during ascent.

The first flow of the Denchai Basalt was apparently the most voluminous, and is chemically transitional to olivine tholeiite. The magma was probably generated by relatively large amounts of partial melting. Small amounts of fractionation of olivine and calcic plagioclase at relatively shallow depths may have yielded the differentiates of flows 2, 3, and 4.

Although linear trends are present between these lowermost four flows and the overlying flows, they are in the opposite sense to typical differentiation trends (Fig. 8). The increasingly alkalic nature of younger flows suggests decreasing amounts of partial melting at possibly greater depths. However relative constancy in incompatible trace element ratios (Table

3) suggests that the magmas of the Denchai Basalt were probably essentially comagmatic [19].

Magmas of flows 5, 6, 7 probably ascended rapidly, carrying spinel lherzolite nodules from depths in the order of 50 km [20]. Presumably the zircon and corundum were also carried, having originated either as cognate megacrysts crystallized at depth or as mantle components.

8. Tectonic setting

The association of alkalic basalts with continental regions of Cenozoic uplift, extensional tectonism, and high heat flow is world-wide (e.g. southwestern U.S.A., southeastern Australia, northern Africa) but the details of the tectonic settings are generally complex [21–23].

During the Cenozoic, Southeast Asia has experienced the formation of deep basins [24,25] and reactivation of faults and lineaments [26], as well as extrusion of alkalic magmas such as the Denchai Basalt, commonly in association with these basins and faults. Many tectonic events may have influenced these activities, including the opening of the Andaman Sea [27,28], rotation of parts of Southeast Asia [29], opening of the South China Sea [30] and interactions between the Indian and Eurasian plates (e.g. [31]). Thus the Late Cenozoic basalts of Southeast Asia are probably an expression of complex tectonic phenomena.

9. Conclusions

The Denchai Basalt consists of seven flows extruded during an interval (or intervals) of reversed polarity of the Earth's magnetic field. The uppermost flow has yielded an age of 5.64 ± 0.28 Ma by the K-Ar method, indicating that the igneous activity occurred during the reversed interval between 6.06 and 5.62 Ma. However the presence of a palaeosol horizon between flows 1 and 7 suggest the possibility that flows 1–6 may have occurred during an older reversed interval. During the igneous activity the chemical compositions of the magmas (apparently extruded from one vent area), changed from "hawaiite" (transitional to continental tholeiite) to true hawaiite and then to basanite. The latter uppermost flow is the probable source of zircon and sapphire mined from placer deposits associated with the basalt. The chemical change is probably the result of decreasing amounts of partial melting rather than a change in source material. The controlling factors in the occurrence of gems appear to be the chemical composition and rate of ascent of the magma.

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> "After many and many a traii
> I was able to spell 'Hawaii'
> Then found that petrographers lack etymological insaiit
> Since they write a trite 'hawaiite' in lieu of the splendid and indubitably right 'HAWAIIITE'."

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